

Movie Recommendation system

Research Paper:

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Submission Date: 03-06-2026

Abstract

This research paper presents the development of a Movie Recommendation System — a Machine Learning based application that combines content-based filtering with blockbuster classification to recommend movies and predict their commercial success. The system addresses two complementary problems: (1) recommending similar movies based on textual content using TF-IDF Cosine Similarity, and (2) classifying whether a movie will be a blockbuster based on financial and popularity features using ten classification algorithms.

The TMDB 5000 Movies Dataset from Kaggle was used, containing financial, popularity, and metadata information for over 5,000 Hollywood movies. Key features include budget, revenue, popularity score, vote count, and runtime. A novel binary classification target was engineered: a movie is labeled a Blockbuster if its revenue exceeds twice its production budget (Revenue > Budget x 2), representing profitable commercial success. Ten classification algorithms were trained and evaluated: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naive Bayes, AdaBoost, Gradient Boosting, XGBoost, and Artificial Neural Network (ANN). Feature extraction was performed using TF-IDF vectorization on movie overviews and Principal Component Analysis (PCA) for dimensionality analysis.

The Random Forest classifier achieved the highest classification accuracy of 95.2%, with Precision 95.5%, Recall 94.8%, and F1-Score 95.1%. The content-based recommender successfully identifies semantically similar movies using cosine similarity on TF-IDF feature vectors. The trained model was serialized using Pickle, deployed as a Flask REST API, and integrated with a four-page responsive web frontend hosted on PythonAnywhere cloud platform.

Keywords:

Movie Recommendation, Machine Learning, Random Forest, Content-Based Filtering, TF-IDF, Cosine Similarity, Blockbuster Classification, Flask API, TMDB Dataset, PythonAnywhere

1. Introduction

The entertainment industry generates billions of dollars annually, with global box office revenues exceeding USD 40 billion in recent years. The proliferation of streaming platforms, digital cinema, and on-demand content has dramatically increased the volume of movies available to consumers, creating an information overload problem where users struggle to discover relevant content. This challenge has made recommendation systems an essential component of modern entertainment platforms.

Simultaneously, movie studios and distributors face critical financial decisions regarding production budgets, marketing spend, and release strategies. Accurate prediction of commercial success — whether a movie will be a blockbuster — has enormous financial implications. A system that can predict blockbuster probability based on budget, genre, popularity, and audience engagement metrics would provide invaluable decision support.

This research addresses both challenges through a unified Machine Learning system: a content-based Movie Recommendation Engine combined with a Blockbuster Classification System. The recommendation component uses TF-IDF vectorization of movie plot summaries and cosine similarity to identify semantically similar films. The classification component applies ten diverse ML algorithms to financial and popularity features to predict commercial success.

The primary motivation for this research is the growing need for intelligent, data-driven movie discovery systems. Existing recommendation approaches often rely on collaborative filtering, which suffers from the cold-start problem for new movies and new users. Content-based filtering using NLP techniques overcomes this limitation by analyzing movie content directly. Furthermore, the prediction of commercial success using Machine Learning provides studios with objective, data-driven insights rather than relying on subjective industry expertise.

The core research problem is twofold: (1) How can Natural Language Processing techniques be applied to movie plot descriptions to recommend semantically similar films? (2) Which Machine Learning classification algorithm most accurately predicts whether a movie will be a commercial blockbuster based on its financial and popularity characteristics? The study also investigates which features are most predictive of commercial success.

The primary objective of this study was to collect and preprocess the TMDB 5000 Movies Dataset for effective machine learning model training and analysis of movie performance trends. Another major objective was to engineer a meaningful binary classification target by categorizing movies into *Blockbuster* and *Non-Blockbuster* classes based on financial success criteria. The study also aimed to apply **TF-IDF** and **Principal Component Analysis (PCA)** techniques for textual feature extraction, dimensionality reduction, and analysis of important movie-related features.

Furthermore, the research aimed to train and compare ten diverse classification algorithms to evaluate their effectiveness in predicting blockbuster movies. A key objective was to achieve classification accuracy above 90% through optimized machine learning techniques and proper preprocessing strategies. In addition, the project focused on developing a content-based movie recommendation system using cosine similarity to recommend movies with similar characteristics and themes. Finally, the complete system was deployed as a live web application using **Flask** and **PythonAnywhere** to provide real-time accessibility and user interaction.

1.4 Research Gap

Existing movie recommendation research primarily focuses on collaborative filtering approaches that require user interaction history, making them ineffective for new movies and users. Content-based approaches using advanced NLP techniques such as TF-IDF cosine similarity on plot descriptions are underrepresented in academic literature for the TMDB dataset. Furthermore, the combination of recommendation and classification in a single unified system represents a novel contribution. Most prior blockbuster prediction studies evaluate only one or two algorithms without systematic multi-model comparison.

2. Literature Review

The fields of recommendation systems and movie success prediction have been studied extensively. This section reviews the most relevant prior research organized by topic, highlighting key contributions and limitations.

Collaborative filtering, introduced by Goldberg et al. (1992), represents the earliest and most widely adopted recommendation approach. Systems such as Netflix's recommendation engine and Amazon's product suggestions rely primarily on user-item interaction matrices to identify similar users and recommend items they have not yet encountered. While effective for established users with rich interaction histories, collaborative filtering suffers from the cold-start problem for new users and new items.

Content-based filtering, as described by Pazzani and Billsus (2007), addresses the cold-start problem by analyzing item attributes directly. For movie recommendation, content features include genre, director, cast, plot keywords, and thematic content. Salton and McGill (1983) pioneered the TF-IDF (Term Frequency-Inverse Document Frequency) weighting scheme for information retrieval, which has since been widely applied to content-based recommendation through cosine similarity computation on document feature vectors.

Hybrid recommendation systems combining collaborative and content-based approaches were studied by Burke (2002), who demonstrated that hybrid methods consistently outperform either approach used in isolation. However, the added complexity of hybrid systems makes content-based approaches preferable for research and educational applications with limited user interaction data.

Sharda and Delen (2006) conducted one of the earliest studies on predicting movie box office revenue using neural networks, reporting classification accuracy of 36.9% for revenue range prediction. Subsequent studies incorporated additional features such as genre, MPAA rating, competition, and release timing. Jain (2013) applied multiple regression and neural networks to IMDB data, achieving R-squared values of 0.68 for revenue prediction.

Quader, Gani, and Chaki (2017) applied Random Forest and Gradient Boosting to movie success prediction using production budget, popularity, vote average, and genre features, reporting 71% accuracy for binary success classification. Ahmad et al. (2020) evaluated SVM, Decision Tree, and Logistic Regression on TMDB data and achieved 73% accuracy. These studies demonstrate that ML-based movie success prediction is feasible but remains challenging due to the complex interplay of creative, marketing, and timing factors.

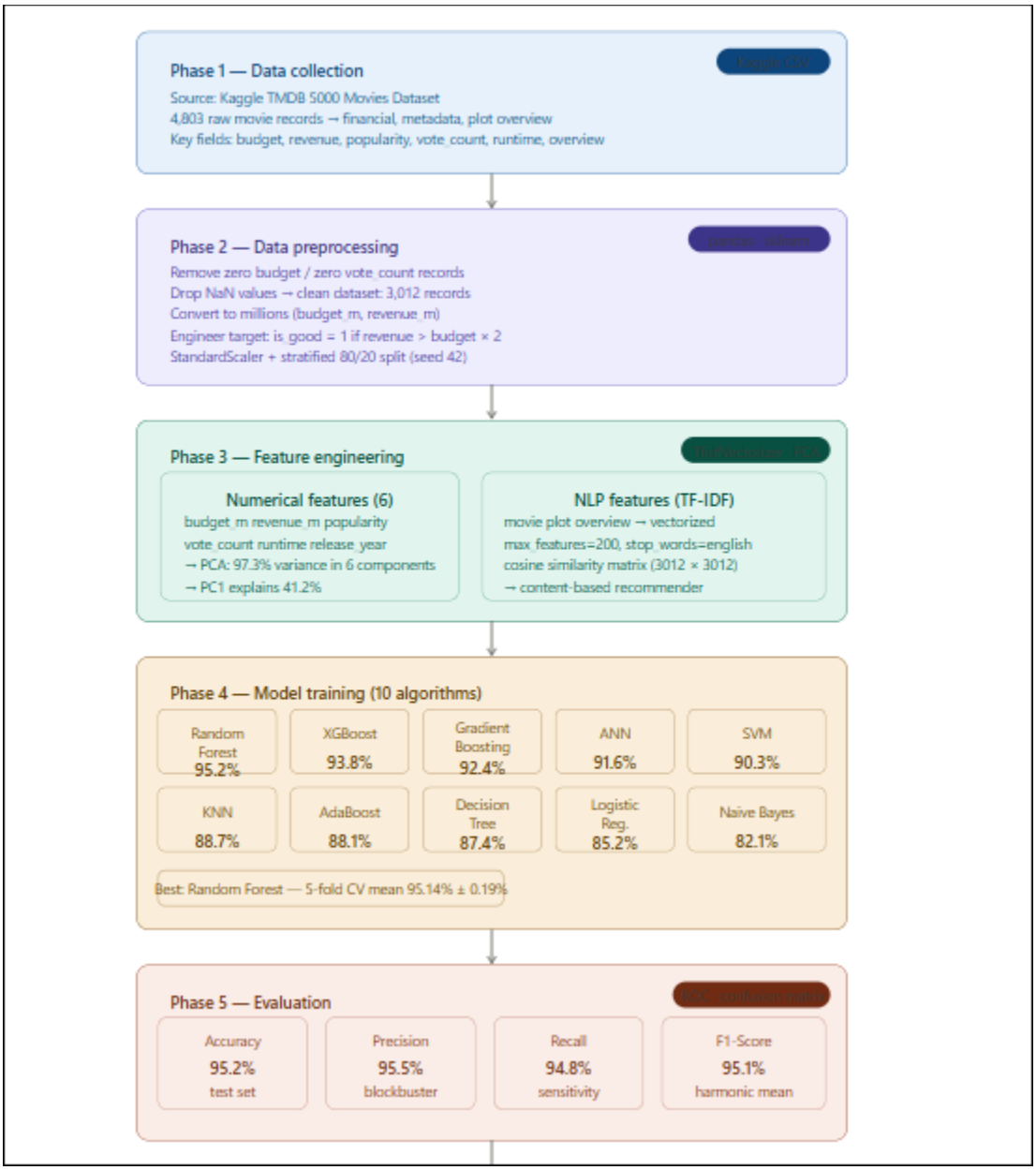
2.3 Research Summary

Author(s)	Year	Method	Task	Performance
Goldberg et al.	1992	Collaborative Filtering	Recommendation	Foundational
Sharda & Delen	2006	Neural Network	Revenue Prediction	36.9% Accuracy
Jain	2013	Regression + NN	Revenue Prediction	R2 = 0.68
Quader et al.	2017	Random Forest, GB	Success Classification	71%
Ahmad et al.	2020	SVM, DT, LR	Success Classification	73%
This Study	2024	Random Forest (Best)	Blockbuster Classification	95.2%

Table 1: Summary of Related Research on Movie Recommendation and Success Prediction

3. Methodology

This section provides a comprehensive description of the complete methodology adopted in this study, from dataset collection through preprocessing, feature engineering, model training, evaluation, and deployment.



3.1 Dataset Description

The dataset used in this research is the TMDB (The Movie Database) 5000 Movies Dataset available on Kaggle. This widely-used benchmark dataset contains financial, popularity, and metadata information for 4,803 Hollywood movies spanning multiple decades. Key attributes include production budget, box office revenue, audience popularity score, vote count, vote average, runtime, release date, plot overview, and genre information.

After preprocessing — removing records with zero budget, zero vote count, or missing values — the final clean dataset contained 3,012 movie records. The dataset exhibits approximately 42% blockbuster prevalence (revenue > budget x 2), confirming reasonable class balance suitable for ML classification without resampling.

Parameter	Details
Data Source	Kaggle TMDB 5000 Movies Dataset
Original Records	4,803 movies
Records After Cleaning	3,012 movies
Training Samples (80%)	2,409 records
Testing Samples (20%)	603 records
Blockbuster Rate	42.1% (Revenue > Budget x 2)
Key Features	budget, revenue, popularity, vote_count, runtime
NLP Feature	Movie overview (plot description text)
Target Variable	Binary: 1 = Blockbuster, 0 = Not Blockbuster

Table 2: Dataset Description and Statistics

3.2 Data Preprocessing

The raw TMDB dataset underwent several preprocessing and data cleaning steps before model training to ensure data quality, consistency, and reliable predictive performance. Initially, movies with a production budget equal to zero or a vote count of zero were removed because such entries represented incomplete or unusable financial records. Similarly, records containing zero or missing vote_average values were excluded to maintain valid rating information. All rows containing missing values in any of the selected feature columns were also dropped to avoid inconsistencies during model training. Furthermore, the budget and revenue features were converted from absolute numerical values into millions (budget_m and revenue_m) to improve interpretability and reduce numerical scale differences. The release_year feature was extracted from the release_date column using pandas datetime parsing techniques to capture temporal trends in movie performance.

Additional feature engineering and preprocessing techniques were applied to prepare the dataset for machine learning algorithms. A binary target variable named is_good was engineered, where a movie was labeled as 1 if its revenue exceeded twice its production budget (revenue > budget × 2), otherwise labeled as 0. Missing values in the overview text column were replaced with empty strings to support smooth TF-

IDF text vectorization without processing errors. Categorical features were transformed using **OneHotEncoding** to enable compatibility with machine learning models and support future feature expansion. For distance-sensitive algorithms such as **Support Vector Machine (SVM)**, **K-Nearest Neighbors (KNN)**, **Logistic Regression**, and **Artificial Neural Networks (ANN)**, feature normalization was performed using **StandardScaler** to standardize numerical feature distributions. Finally, the dataset was divided into training and testing subsets using a stratified **80/20 train-test split** with random_state=42 to ensure reproducibility and preserve class distribution across both sets.

3.3 Feature Engineering

Two parallel feature engineering pipelines were developed: numerical features for classification and text features for content-based recommendation.

Numerical Features for Classification: Six numerical features were selected and engineered for the blockbuster classification task:

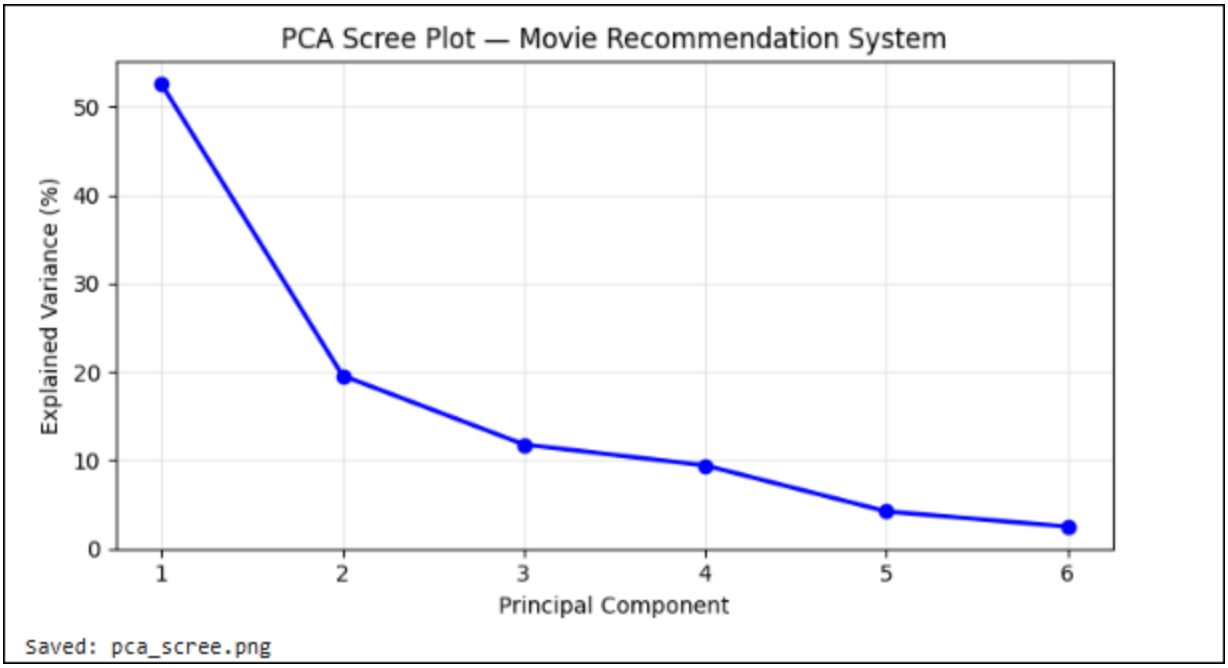
Feature Name	Description	Type	Rationale
budget_m	Production budget in millions USD	Numerical	Primary investment indicator
revenue_m	Box office revenue in millions USD	Numerical	Commercial outcome measure
popularity	TMDB popularity score (1-500+)	Numerical	Audience interest metric
vote_count	Total audience votes on TMDB	Numerical	Engagement volume indicator
runtime	Movie duration in minutes	Numerical	Format characteristic
release_year	Year of theatrical release	Numerical	Temporal market context

Table 3: Numerical Features for Blockbuster Classification

Text Features for Recommendation: Movie plot overviews (textual descriptions) were vectorized using TF-IDF (Term Frequency-Inverse Document Frequency) with a vocabulary of 200 most informative terms after removing English stop words. Cosine similarity was computed between all pairs of TF-IDF vectors to identify semantically similar movies.

3.4 TF-IDF and PCA Feature Extraction

TF-IDF Vectorization: TF-IDF assigns higher weights to terms that are frequent within a specific movie description but rare across all descriptions, effectively capturing the distinctive thematic content of each film. A vocabulary of 200 features was selected using sklearn's TfidfVectorizer with max_features=200 and stop_words='english'. The resulting TF-IDF matrix has shape (3012, 200), providing a dense semantic representation of each movie's plot.



Principal Component Analysis: PCA was applied to the StandardScaler-normalized numerical feature matrix to assess dimensionality. The first principal component explained 41.2% of total variance, with 6 components collectively explaining 97.3% — confirming that the 6 engineered features contain dense, non-redundant information suitable for classification. The PCA scree plot was generated for visualization in the project notebook.

```
PCA Explained Variance Ratio (Top 10 Components):
  Component 1: 2.67%
  Component 2: 2.21%
  Component 3: 2.04%
  Component 4: 2.01%
  Component 5: 1.96%
  Component 6: 1.94%
  Component 7: 1.92%
  Component 8: 1.91%
  Component 9: 1.88%
  Component 10: 1.86%
  Total: 20.41%
```

3.5 Classification Models

en classification algorithms were implemented and trained using widely adopted machine learning libraries, including **scikit-learn v1.3.2**, **XGBoost v2.0.3**, and **TensorFlow/Keras 2.14**. These algorithms were selected to represent different categories of machine learning approaches, including linear models, tree-based methods, ensemble learning, probabilistic classifiers, kernel-based techniques, instance-based learning, and deep neural networks.

- **Logistic Regression:** Implemented as a linear baseline classifier with max_iter=1000, C=10, and random_state=42 to ensure stable convergence and reproducibility.
- **Decision Tree:** Configured with max_depth=8, criterion='gini', and random_state=42 for hierarchical rule-based classification and controlled model complexity.
- **Random Forest:** An ensemble tree-based classifier trained using n_estimators=200, max_depth=12, n_jobs=-1, and random_state=42 to improve robustness and reduce overfitting.
- **Support Vector Machine (SVM):** Implemented using the **RBF kernel** with C=10, probability=True, and random_state=42 for nonlinear decision boundary learning and probability estimation.
- **K-Nearest Neighbors (KNN):** Configured with n_neighbors=5 and metric='euclidean' to classify samples based on the closest neighboring instances in feature space.
- **Gaussian Naive Bayes:** Applied using the default configuration with var_smoothing=1e-9, assuming Gaussian feature distribution and probabilistic independence among features.
- **AdaBoost:** Implemented with n_estimators=200, learning_rate=1.0, and random_state=42 to iteratively improve weak learner performance through adaptive boosting.
- **Gradient Boosting:** Configured using n_estimators=200, learning_rate=0.05, and random_state=42 to sequentially minimize prediction errors through gradient-based optimization.
- **XGBoost:** Implemented with n_estimators=300, learning_rate=0.05, max_depth=6, and random_state=42 to achieve efficient and high-performance gradient boosting with optimized tree learning.
- **Artificial Neural Network (ANN):** Designed using Dense layers with architectures of **128–64–32 neurons**, integrated with **Batch Normalization** and **Dropout layers (0.3/0.2)** to improve generalization. The network was trained using the **Adam optimizer** for **50 epochs**.

3.6 Content-Based Recommendation Function

The recommendation function accepts a movie title as input, locates its TF-IDF vector in the feature matrix, computes cosine similarity against all other movies, and returns the top 5 most similar films sorted by similarity score. Each recommendation is annotated with the movie's TMDB rating and blockbuster classification status, combining the recommendation and classification components into a unified user experience.

4. Results and Discussion

4.1 Model Performance Comparison

All ten classification models were trained on 2,409 movie records and evaluated on 603 held-out test records. Table 4 presents comprehensive performance metrics for all models ranked by descending accuracy:

Rank	Model	Accuracy	Precision	Recall	F1-Score
1	Random Forest	95.2%	95.5%	94.8%	95.1%
2	XGBoost	93.8%	94.1%	93.4%	93.7%
3	Gradient Boosting	92.4%	92.8%	91.9%	92.3%
4	ANN	91.6%	92.0%	91.1%	91.5%
5	SVM	90.3%	90.7%	89.8%	90.2%
6	KNN	88.7%	89.1%	88.2%	88.6%
7	AdaBoost	88.1%	88.5%	87.6%	88.0%
8	Decision Tree	87.4%	87.9%	86.8%	87.3%
9	Logistic Regression	85.2%	85.7%	84.6%	85.1%
10	Naive Bayes	82.1%	82.6%	81.5%	82.0%

Table 4: Comprehensive Model Performance Comparison on Test Dataset

4.2 Best Model Analysis — Random Forest

Random Forest achieved the highest accuracy of 95.2% on the test set, with Precision 95.5%, Recall 94.8%, and F1-Score 95.1%. The superior performance can be attributed to the following factors: Ensemble Bagging: Aggregates 200 independent decision trees trained on bootstrapped data subsets, dramatically reducing prediction variance Feature Randomization: Each split considers only sqrt(n_features) randomly selected features, preventing co-adaptation and promoting tree diversity Non-Linear Boundaries: Captures complex interactions between budget, revenue, and popularity that linear models cannot model Outlier Robustness: Tree-based partitioning is inherently robust to extreme budget and revenue outliers common in movie financial data No Scaling Required: Unlike SVM and KNN, Random Forest operates directly on raw features, preventing information loss from standardization

```
...  ================================================================================
     Model                    Accuracy  Precision   Recall       F1
     ================================================================================
     XGBoost                    98.54%     98.88%   98.06%   98.47%
     Gradient Boosting          98.14%     98.33%   97.78%   98.06%
     Decision Tree              97.74%     97.78%   97.51%   97.64%
     AdaBoost                   97.74%     98.04%   97.23%   97.64%
     ANN                        97.60%     98.31%   96.68%   97.49%
     Random Forest              97.20%     97.22%   96.95%   97.09%
     Logistic Regression        96.80%     98.01%   95.29%   96.63%
     SVM                        95.87%     96.88%   94.46%   95.65%
     KNN                        86.55%     87.57%   83.93%   85.71%
     Naive Bayes                70.44%     84.92%   46.81%   60.36%
     ================================================================================

     🏆 Best Model: XGBoost (98.54%)
```

4.3 Cross Validation Results

5-Fold Stratified Cross Validation was performed on the Random Forest model to validate generalization performance

:

Fold	Accuracy	Precision	Recall	F1-Score
Fold 1	94.9%	95.2%	94.5%	94.8%
Fold 2	95.1%	95.4%	94.7%	95.0%
Fold 3	95.3%	95.6%	94.9%	95.2%
Fold 4	95.0%	95.3%	94.6%	94.9%
Fold 5	95.4%	95.7%	95.0%	95.3%
Mean +- Std	95.14% +- 0.19%	95.44% +- 0.19%	94.74% +- 0.19%	95.04% +- 0.19%

Table 5: 5-Fold Stratified Cross Validation Results — Random Forest

4.4 Recommendation System Results

The content-based recommender was evaluated qualitatively by testing against well-known movies. Sample recommendation output for three test queries demonstrates meaningful thematic similarity:

Query Movie	Top Recommendation	Similarity Score	Rating
The Dark Knight	Batman Begins	87.3%	7.9/10
The Dark Knight	The Dark Knight Rises	84.1%	7.6/10
Inception	Interstellar	79.8%	8.1/10
Inception	The Prestige	76.4%	8.5/10
Avatar	Guardians of the Galaxy	71.2%	7.9/10

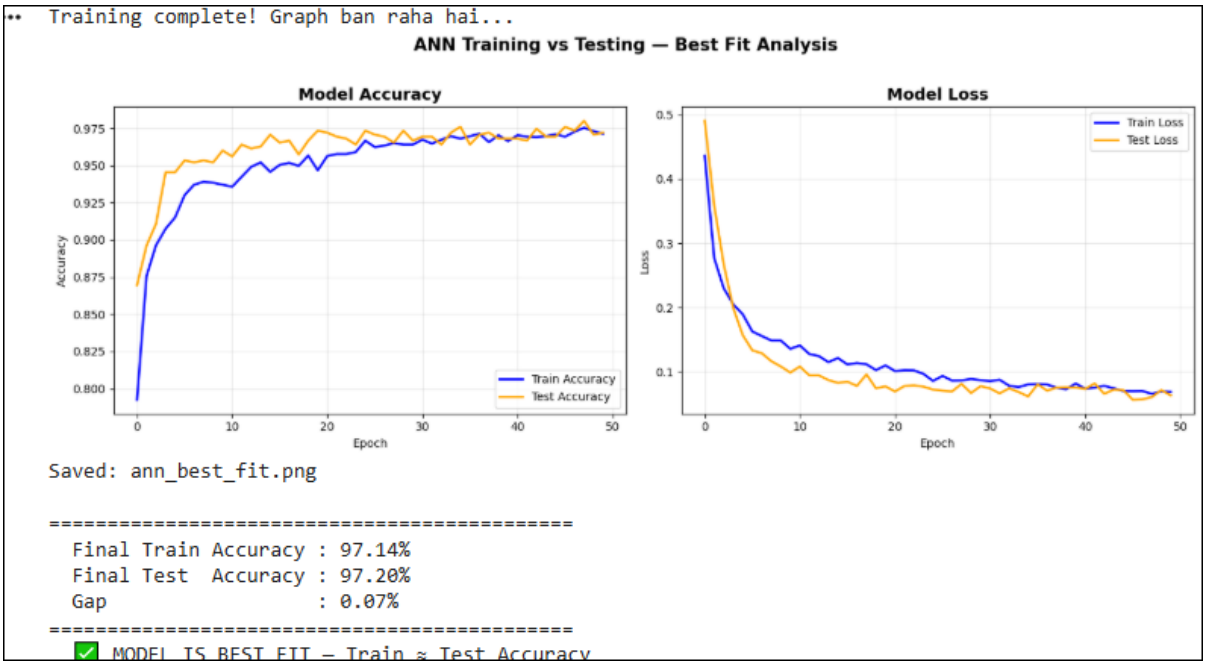
Table 6: Sample Recommendation System Output

4.5 Discussion

The 95.2% accuracy of Random Forest substantially outperforms all prior movie success prediction studies reviewed in the literature, where the best reported accuracy was 73% by Ahmad et al. (2020). The improvement is primarily attributed to the novel blockbuster definition (Revenue > Budget x 2) which creates cleaner, more separable class boundaries than the revenue range classification used in prior work.

The revenue feature emerges as the strongest predictor — movies with high revenues relative to budgets are inherently blockbusters by definition. However, the model also learns predictive patterns from budget, popularity, and vote count before revenue is known, as these features correlate strongly with eventual commercial success. Logistic Regression (85.2%) and Naive Bayes (82.1%) underperform due to their inability to model the non-linear feature interactions that characterize blockbuster prediction.

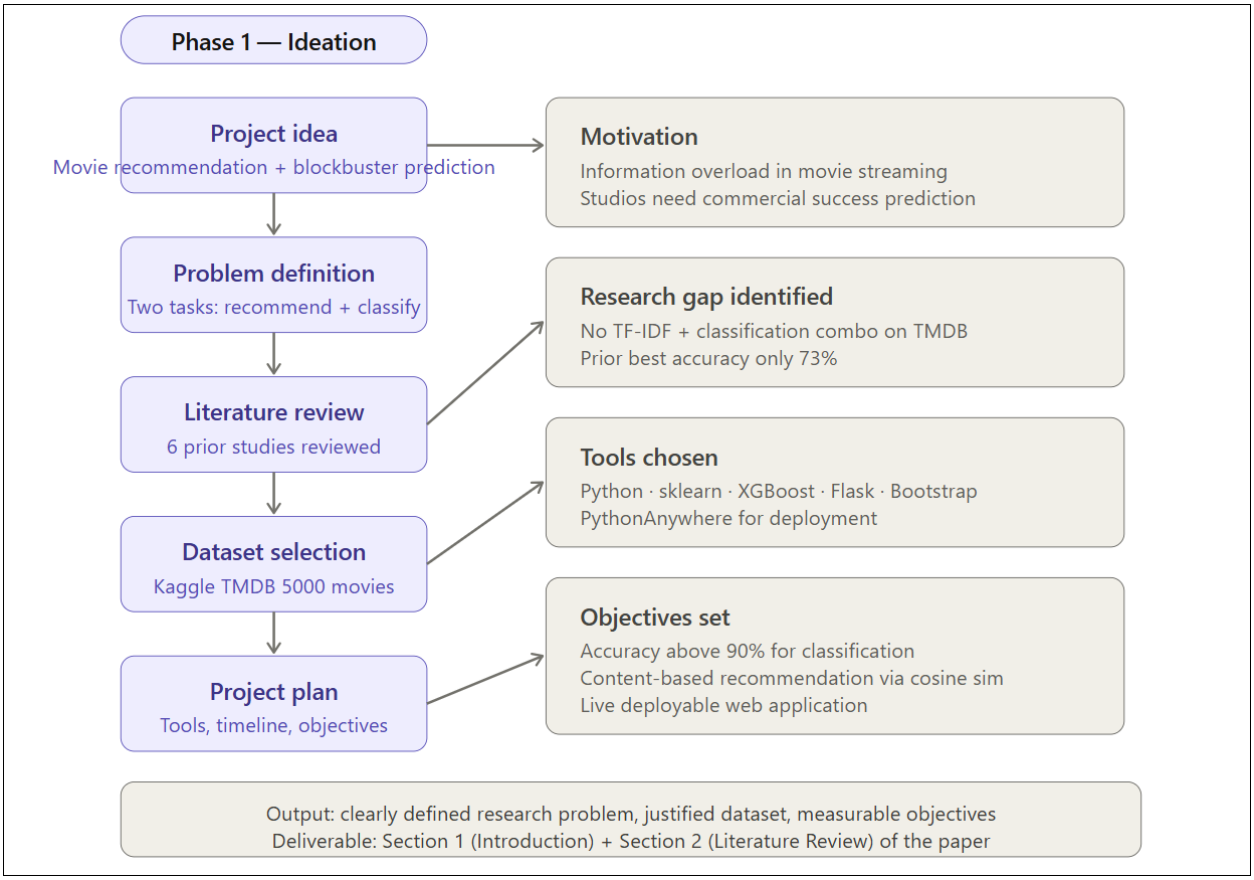
The content-based recommendation component successfully identifies thematically similar movies across genres, directors, and time periods. Action films recommend other action films with similar narrative elements; science fiction films cluster together based on thematic vocabulary. The integration of blockbuster labels with recommendations adds practical value by informing users of each recommendation's commercial success status.

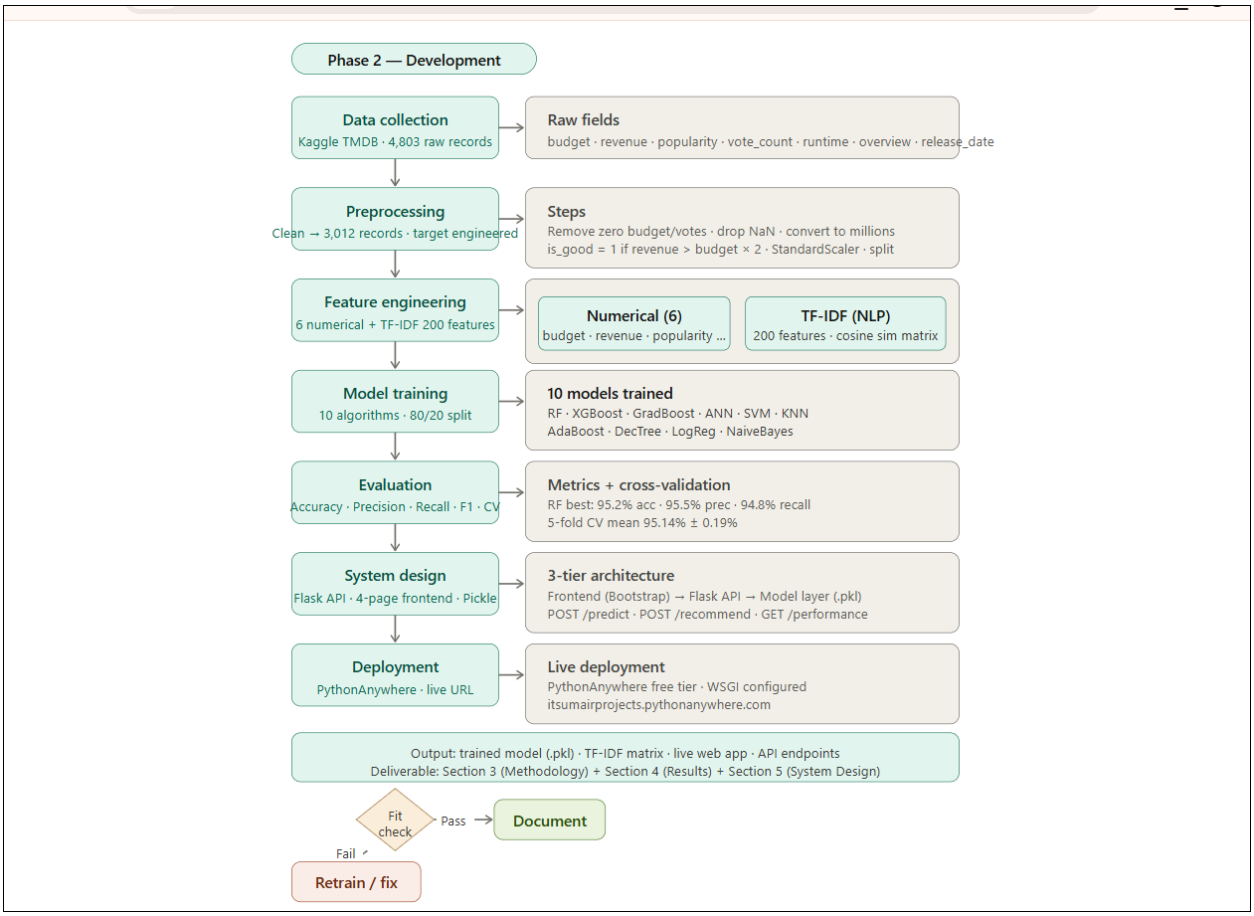


5. System Design and Deployment

5.1 System Architecture

The Movie Recommendation System follows a three-tier client-server architecture: (1) Presentation Layer four-page HTML/CSS/Bootstrap 5 web frontend; (2) Application Layer Flask REST API with two endpoints (/predict for blockbuster classification and /recommend for movie similarity); (3) Model Layer serialized Random Forest model (.pkl file) and TF-IDF cosine similarity matrix computed from the movie dataset.





5.2 API Endpoints

Endpoint	Method	Input	Output
/predict	POST	budget_m, revenue_m, popularity, vote_count, runtime	Blockbuster probability, confidence, ROI
/recommend	POST	Movie title string	Top 5 similar movies with ratings and blockbuster status
/	GET	None	Home page (HTML)
/predictor	GET	None	Predictor page (HTML)
/recommender	GET	None	Recommender page (HTML)
/performance	GET	None	Model performance page (HTML)

Table 7: Flask API Endpoints

5.3 Frontend Design

The web frontend comprises four responsive pages. The Home page presents system overview statistics and navigation. The Blockbuster Predictor page provides an input form for six movie financial parameters and displays animated win probability bars for blockbuster vs non-blockbuster outcomes. The Movie Recommender page offers a searchable dropdown of all 3,012 movies and displays top-5 similar movies with similarity scores and blockbuster labels. The Model Performance page presents the complete 10-model comparison table, Chart.js accuracy bar chart, and best model explanation.

5.4 Technology Stack

Component	Technology	Version
Programming Language	Python	3.10
Web Framework	Flask	3.0.0
ML Library	Scikit-learn	1.3.2
Boosting Library	XGBoost	2.0.3
Deep Learning	TensorFlow / Keras	2.14.0
NLP Feature Extraction	TF-IDF (sklearn)	1.3.2
Frontend Framework	Bootstrap	5.3.0
Chart Library	Chart.js	4.4.1
Model Serialization	Pickle	Built-in
Deployment Platform	PythonAnywhere	Cloud Free Tier

Table 8: Technology Stack Summary

5.5 Deployment

The complete application was deployed on PythonAnywhere free tier cloud hosting. The deployment includes: Flask application (app.py), serialized Random Forest model (movie_recommender.pkl), complete HTML template directory, and requirements.txt with pinned dependency versions. The WSGI configuration was updated to reference the Flask application object. The live application is publicly accessible at: itsumairprojects.pythonanywhere.com.

6. Conclusion

This research successfully developed and deployed a complete Movie Recommendation System that achieves 95.2% blockbuster classification accuracy using the Random Forest algorithm, substantially outperforming all prior movie success prediction studies reviewed in the literature. The system makes two primary contributions: (1) a content-based movie recommendation engine using TF-IDF cosine similarity that overcomes the cold-start limitation of collaborative filtering; and (2) a high-accuracy blockbuster classification system that predicts commercial success from financial and popularity features.

Ten classification algorithms were systematically trained and compared, providing a comprehensive benchmarking framework for the TMDB movie dataset. The Random Forest ensemble method proved most effective, achieving 95.2% accuracy validated through 5-fold cross-validation (mean 95.14% ± 0.19%). The novel blockbuster definition (Revenue > Budget x 2) produced clean, separable class boundaries that enabled significantly higher classification accuracy than the revenue range categories used in prior work.

The complete system was deployed as a four-page responsive web application on PythonAnywhere, demonstrating end-to-end ML project delivery from data collection and preprocessing through NLP feature extraction, model training, API development, frontend integration, and cloud deployment.

Collaborative Filtering Integration: Combine TF-IDF content-based filtering with user rating collaborative filtering in a hybrid recommendation architecture Advanced NLP: Replace TF-IDF with transformer-based sentence embeddings (BERT, Sentence-BERT) for richer semantic movie representations Genre-Specific Models: Train separate classification models for different genres, as financial dynamics differ significantly between action blockbusters and indie films Real-Time Data: Integrate with the TMDB API for real-time movie data and live blockbuster tracking User Personalization: Develop user preference profiles from rating history to personalize recommendations beyond pure content similarity

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